U.S.N.										

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

December 2025 / January 2026 Semester End Main Examinations

Programme: B.E. **Semester: VI**
Branch: Computer Science and Engineering **Duration: 3 hrs.**
Course Code: 23CS6PCMAL **Max Marks: 100**
Course: MACHINE LEARNING

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

		UNIT - I	***CO***	***PO***	**Marks**
1	a)	Analyze when machine learning is the appropriate solution to a problem. Give examples.	*CO2*	*PO2*	**05**
	b)	Explain the concepts of batch learning and online learning	*CO1*	*PO1*	**05**
	c)	You are asked to build a spam email detector. Outline the end-to-end approach, including data preparation, feature engineering, model selection, and evaluation metrics.	*CO2*	*PO2*	**10**
		OR			
2	a)	Analyze the difference between instance-based and model-based learning with examples?	*CO2*	*PO2*	**05**
	b)	Explain the differences between supervised, unsupervised, and reinforcement learning with suitable examples.	*CO1*	*PO1*	**05**
	c)	Explain the main challenges involved in building and deploying machine learning models.	*CO1*	*PO1*	**10**
		UNIT - II			
3	a)	i. Consider the dataset below to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether they studied.	*CO3*	*PO3*	**10**

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

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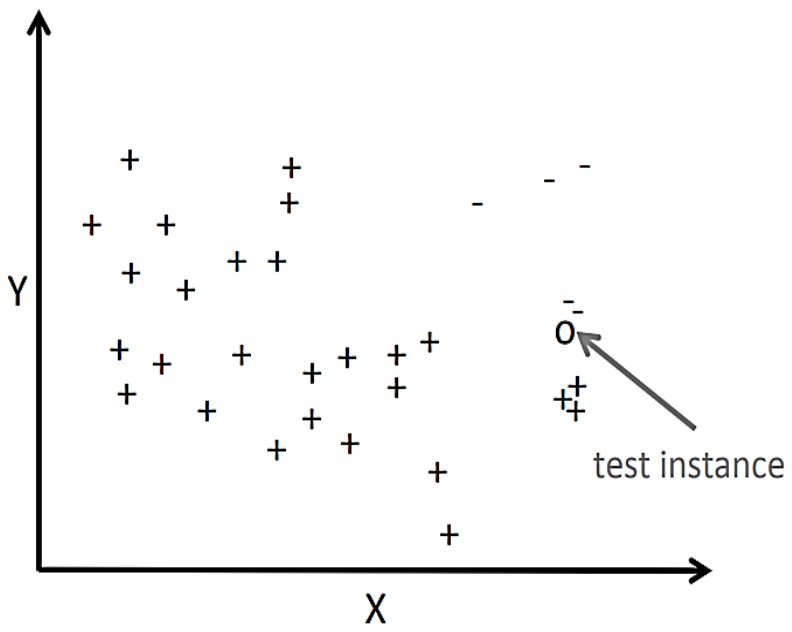
			• What is the entropy H(Passed)? • What is the entropy H(Passed | GPA)? ii. Explain the role of entropy and information gain in building a decision tree?			
		b)	Compare and contrast Linear Regression and Logistic Regression. In your answer, explain the following: • The type of problems each algorithm is suited for • The mathematical models used in each • The output produced by each • Limitations and when one is preferred over the other • Use one real-world example for each to illustrate your points	*CO2*	*PO2*	**10**
			OR			
	4	a)	Explain the working principle of Support Vector Machines (SVM) for both linearly and non-linearly separable data. How do kernel functions help in handling non-linear data? Illustrate with suitable diagrams or examples.	*CO1*	*PO1*	**10**
		b)	i. Suppose we have the following training set of positive (+) and negative (-) instances and a single test instance (o). All instances are projected onto a vector space of two real-valued features (X and Y). Answer the following questions. Assume "unweighted" KNN (every nearest neighbor contributes equally to the final vote).  • What would be the class assigned to this test instance for K=1 • What would be the class assigned to this test instance for K=3 • What would be the class assigned to this test instance for K=5	*CO2*	*PO2*	**10**

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			• Setting K to a large value seems like a good idea. We get more votes! Given this particular training set, would you recommend setting K = 11? Why or why not? ii. Write an algorithm for K-Nearest Neighbor Classification			
			UNIT - III			
	5	a)	i. Define a paired t-test. ii. When is it used in model comparison? What assumptions must be satisfied to use a paired t-test for comparing two classification models? iii. Given two models A and B and its performance. Perform a paired t-test at $\alpha = 0.05$. Model A accuracy over 5 folds: 90, 85, 88, 92, 87 Model B accuracy over same folds: 88, 83, 87, 91, 85	*CO2*	*PO2*	**10**
		b)	Define Recall metric. Why is it critical in scenarios like disease detection? If a cancer detection model has low recall, what are the consequences? Given TP = 40, FN = 60. Calculate recall.	*CO2*	*PO2*	**10**
			OR			
	6	a)	i. Define stratified k-fold cross-validation. How is it different from standard k-fold cross-validation? ii. In a binary classification with imbalanced data (90% class A, 10% class B), which cross-validation strategy is preferred and why? iii. If you perform 4-fold stratified cross-validation on a dataset with 800 samples (720 of Class 0 and 80 of Class 1), how many Class 1 samples will be in each fold?	*CO2*	*PO2*	**10**
		b)	What does the ROC curve represent? What do the axes stand for? Why is a ROC curve more informative than just using accuracy? A model has TPR = 0.85 and FPR = 0.15 at a certain threshold. Plot the point on a ROC curve and describe the model performance.	*CO1*	*PO1*	**10**
			UNIT - IV			
	7	a)	i. Describe the K-means clustering algorithm. ii. Apply K(=2)-Means algorithm over the data (185, 72), (170, 56), (168, 60), (179,68), (182,72), (188,77) up to two iterations and show the clusters. Initially choose first two objects as initial centroids.	*CO3*	*PO3*	**10**
		b)	A dataset contains 200 features, and your model is overfitting. How can PCA help? Describe the process of applying PCA and how to decide the number of components to retain.	*CO2*	*PO2*	**10**
			OR			
	8	a)	Explain the working of the Random Forest algorithm. How does it reduce overfitting compared to a single decision tree? Analyze its advantages, limitations, and suitable application scenarios.	*CO1*	*PO1*	**10**

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		b)	What is Ensemble Learning? Describe the working principles of Bagging and Boosting. Compare their impact on bias and variance, and explain which scenarios are best suited for each method.	*CO1*	*PO1*	**10**
			UNIT - V			
	9	a)	A warehouse robot is trained using reinforcement learning to pick up and deliver packages to designated locations while avoiding obstacles and other moving robots. Identify the possible: (i) States (ii) Actions (iii) Reward function design	*CO2*	*PO2*	**05**
		b)	Describe Q-Learning in reinforcement learning. Explain its working principle and the update rule. Illustrate the Q-learning process with an example of a simple grid world environment.	*CO1*	*PO1*	**10**
		c)	Briefly explain the role of generalization in reinforcement learning.	*CO1*	*PO1*	**05**
			OR			
	10	a)	A vending machine dispenses either snacks or drinks, depending on the user's choice. However, due to a mechanical fault, the machine sometimes gives the wrong item. If the user selects Snack (S): 80% chance of getting a Snack, 20% chance of getting a Drink. If the user selects Drink (D): 90% chance of getting a Drink, 10% chance of getting a Snack. The rewards are: • Getting the correct item: +10 • Getting the wrong item: –5 Design a decision-making system for this vending machine using a Markov Decision Process (MDP)	*CO3*	*PO3*	**05**
		b)	Define active reinforcement learning. How does it differ from passive reinforcement learning?	*CO1*	*PO1*	**05**
		c)	Explain the trade-off between exploration and exploitation in reinforcement learning. How does the ε-greedy policy manage this trade-off?	*CO1*	*PO1*	**10**

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